

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 May/June 2021 Worked Solutions

May/June 2021 paper rebuilt with the worked solution after each question.

9

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P2

PAST PAPER

WMA12/01 May/June 2021

May/June 2021 | 9 questions | 75 marks

9

questions

75

marks

Answers

worked solution
after each
question

Question 1

Arithmetic Sequences

1. Adina is saving money to buy a new computer. She saves £5 in week 1, £5.25 in week 2, £5.50 in week 3 and so on until she has enough money, in total, to buy the computer.

She decides to model her savings using either an arithmetic series or a geometric series.

Using the information given,

- (a) (i) state with a reason whether an arithmetic series or a geometric series should be used,
- (ii) write down an expression, in terms of n , for the amount, in pounds (£), saved in week n .

(3)

Given that the computer Adina wants to buy costs £350

- (b) find the number of weeks it will take for Adina to save enough money to buy the computer.

(4)

(Total 7 marks)

Leave
blank

Worked Solution - Question 1

1. Use the arithmetic formula

Identify the first term, common difference, and required term or sum from the question.

2. Substitute and solve

Use the arithmetic term or sum formula, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: + So 37 weeks

Final answer

+ So 37 weeks

Question 2

Sequences & Series

2.



Figure 1

Figure 1 shows a sketch of the curve with equation $y = 4^x$

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(a) On Diagram 1, sketch the curve with equation

(i) $y = 2^x$

(ii) $y = 4^x - 6$

Label clearly the coordinates of any points of intersection with the coordinate axes.

(4)

The curve with equation $y = 2^x$ meets the curve with equation $y = 4^x - 6$ at the point P .

(b) Using algebra, find the exact coordinates of P .

(4)

Question 2 continued



Diagram 1

(Total 8 marks)

Worked Solution - Question 2

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: therefore $- = 2 \log 3 x = oe$

Final answer

therefore $- = 2 \log 3 x = oe$

Question 3

Proof

3. (i) Prove that for all single digit prime numbers, p ,

$$p^3 + p \text{ is a multiple of 10}$$

(2)

- (ii) Show, using algebra, that for $n \in \mathbb{N}$

$$(n + 1)^3 - n^3 \text{ is not a multiple of 3}$$

(3)

(Total 5 marks)

Worked Solution - Question 3

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: The required statement has been proved.

Final answer

The required statement has been proved.

Question 4

Binomial Expansion

4. (a) Find, in ascending powers of x , up to and including the term in x^3 , the binomial expansion of

$$\left(2 + \frac{x}{8}\right)^{13}$$

fully simplifying each coefficient.

(4)

- (b) Use the answer to part (a) to find an approximation for 2.0125^{13}

Give your answer to 3 decimal places.

(3)

Without calculating 2.0125^{13}

- (c) state, with a reason, whether the answer to part (b) is an overestimate or an underestimate.

(1)

Leave
blank

(Total 8 marks)

Worked Solution - Question 4

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 5

Integration

5.

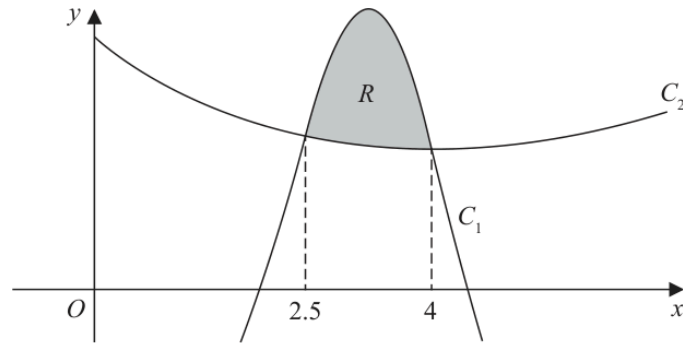


Figure 2

Figure 2 shows a sketch of part of the graph of the curves C_1 and C_2

The curves intersect when $x = 2.5$ and when $x = 4$

A table of values for some points on the curve C_1 is shown below, with y values given to 3 decimal places as appropriate.

x	2.5	2.75	3	3.25	3.5	3.75	4
y	5.453	7.764	9.375	9.964	9.367	7.626	5

Using the trapezium rule with all the values of y in the table,

- (a) find, to 2 decimal places, an estimate for the area bounded by the curve C_1 , the line with equation $x = 2.5$, the x -axis and the line with equation $x = 4$

(4)

The curve C_2 has equation

$$y = x^{\frac{3}{2}} - 3x + 9 \quad x > 0$$

- (b) Find $\int \left(x^{\frac{3}{2}} - 3x + 9 \right) dx$

(3)

The region R , shown shaded in Figure 2, is bounded by the curves C_1 and C_2

- (c) Use the answers to part (a) and part (b) to find, to one decimal place, an estimate for the area of the region R .

(3)

(Total 10 marks)

Worked Solution - Question 5

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: ... =

Final answer

... =

Question 6

Circles

6. A circle has equation

$$x^2 - 6x + y^2 + 8y + k = 0$$

where k is a positive constant.

Given that the x -axis is a tangent to this circle,

- (a) find the value of k .

(3)

The circle meets the coordinate axes at the points R , S and T .

- (b) Find the exact area of the triangle RST .

(4)

(Total 7 marks)

Worked Solution - Question 6

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: $+ =$ or $"4" "16" "9" y +/-$
 $= - -$ Or

Final answer

$+ =$ or $"4" "16" "9" y +/-$ $= - -$ Or

Question 7

Laws of Logarithms

7. (a) Given that

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25)$$

show that

$$2x^3 - 3x^2 - 30x + 56 = 0$$

(4)

- (b) Show that
- -4
- is a root of this cubic equation.

(2)

- (c) Hence, using algebra and showing each step of your working, solve

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25)$$

(4)

(Total 10 marks)

Worked Solution - Question 7

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: - - = (Equation not defined for $x = -4$ so) solutions are 2 and 7 2

Final answer

- - = (Equation not defined for $x = -4$ so) solutions are 2 and 7 2

Question 8

Trigonometric Equations

or
In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < \theta < 360^\circ$, the equation

$$3 \sin(\theta + 30^\circ) = 7 \cos(\theta + 30^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) (a) Show that the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

can be written in the form

$$\sin x (a \cos^2 x + b \cos x + c) = 0$$

where a , b and c are constants to be found.

- (b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

(6)

(Total 10 marks)

Worked Solution - Question 8

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: = +/- = Both of x =

Final answer

= +/- = Both of x =

Question 9

Integration

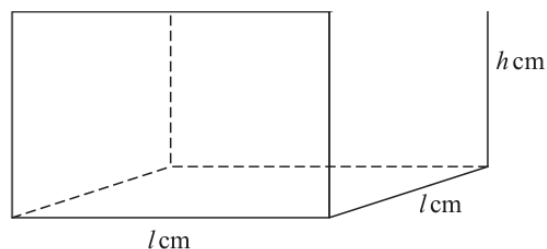


Figure 3

Figure 3 shows a sketch of a square based, open top box.

The height of the box is h cm, and the base edges each have length l cm.

Given that the volume of the box is $250\,000\text{ cm}^3$

- (a) show that the external surface area, $S\text{ cm}^2$, of the box is given by

$$S = \frac{250\,000}{h} + 2000\sqrt{h} \quad (3)$$

- (b) Use algebraic differentiation to show that S has a stationary point when $h = 250^k$ where k is a rational constant to be found.

(5)

- (c) Justify by further differentiation that this value of h gives the minimum external surface area of the box.

(2)

Q9

(Total 10 marks)

(Total 10 marks)

Worked Solution - Question 9

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $= >$ hence gives the minimum value

Final answer

$= >$ hence gives the minimum value